

Numerical and Approximate Solution Methods for the Signature (4,4) Einstein-Spinor Components on $\{\mathbf{x}_0, \dots, \mathbf{x}_7\}$

Complete solver-method provenance and dark-sector interpretation report

Reproducible exact-real implementation

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Abstract

This report gives the numerical strategy for solving the full 18-equation homogeneous reduction of the coordinate-explicit Einstein-spinor model written in

`coordinates = {x0, x1, x2, x3, x4, x5, x6, x7}`

with evolution variable `x4`. The method is designed to solve each component equation in one coupled stiff system, quantify approximation error, and expose model-internal dark-energy-like and dark-matter-like behavior from the 16-component commuting spinor condensate.

1 System solved and solver target

State vector:

$$y(x4) = (a, H, \psi_1, \dots, \psi_{16}) \in \mathbb{R}^{18}.$$

The solved component equations are:

$$\begin{aligned} \frac{da}{dx4} &= aH, & \frac{dH}{dx4} &= -\frac{\kappa_8}{6}(\rho + p), \\ \frac{d\psi_A}{dx4} &= -\frac{7}{2}H\psi_A - V_S \sum_{B=1}^{16} (\gamma^4)_{AB}\psi_B, & A &= 1, \dots, 16. \end{aligned}$$

Auxiliary closures:

$$\begin{aligned} S &= \psi^\top C \psi, & \rho &= \frac{1}{20}S + \frac{19}{20}S^{1/5}, \\ p &= -\frac{19}{25}S^{1/5}, & V_S &= \frac{1}{20} + \frac{19}{100}S^{-4/5}. \end{aligned}$$

2 Best numerical method used for all component equations

The best method for this system, as implemented and verified in this repository, is variable-step variable-order BDF with Newton iterations via the pinned pure- Rust SUNDIALS/CVODE engine.

Reasoning for this choice:

1. The coupled Einstein-spinor equations are mildly stiff near the transition where dilution and self-interaction contributions rebalance.
2. The spinor block is linear in `psi` for fixed `(H,S)` but nonlinear through `S(psi)`, which benefits from implicit stability.

3. BDF + Newton gives robust long-interval integration with deterministic tolerances and reproducible step histories under fixed binaries and inputs.

Canonical solver settings:

- Interval: `x4` in `[-0.2, 1.5]`.
- Output step: `0.01`.
- Relative tolerance: `1e-11`.
- Absolute tolerance: `1e-13`.
- Maximum internal step: `0.002`.

Canonical run summary:

- Samples: `171`.
- Internal solver steps: `1372`.
- RHS evaluations: `1486`.
- Acceleration transition at `x4=-0.138005293303244986`.

3 Approximation and error-control method

The approximation strategy is fully controlled, not ad hoc:

1. Integrate the full coupled ODE once with canonical tolerances.
2. Repeat integration with identical settings to enforce byte-level replay.
3. Integrate a refined run with tighter controls (`rtol=1e-12`, `atol=1e-14`, `max_step=0.001`).
4. Compare canonical and refined states componentwise.
5. Recompute every derived quantity from raw state output independently.

Independent consistency checks applied to each equation family:

1. Friedmann closure check from the `44` Einstein component.
2. Pressure/acceleration consistency from the spatial Einstein components.
3. Five-point finite-difference residual checks on all `18` ODE components.
4. Condensate power-law consistency:

$$S(x4) \sim a(x4)^{-7}.$$

1. Deterministic replay byte-identity for history and summary artifacts.

Canonical maximum relative errors from independent checker:

$$\varepsilon_S \leq 7.44192091691309166 \times 10^{-9},$$

$$\varepsilon_\rho \leq 1.48948308787786630 \times 10^{-9},$$

$$\varepsilon_F \leq 1.20354325050525201 \times 10^{-9}.$$

These bounds demonstrate that the approximate numerical solution tracks the component equations at high precision over the full interval.

4 Equation-by-equation numerical treatment

Each equation family is solved as part of the same implicit coupled system:

1. **a** equation: treated as a linear transport equation once **H** is known per implicit stage.
2. **H** equation: nonlinear scalar equation coupled through **rho(S)** and **p(S)**.
3. **psi_A** equations (**A=1,...,16**): linear matrix action in **psi** with nonlinear coefficient **V_S(S(psi))**.

No equation is decoupled or solved by shortcut substitution; all 18 components are advanced simultaneously in each implicit Newton solve.

5 Dark-energy-like and dark-matter-like provenance from component output

The model's possible dark-sector correspondences are fully internal and conditional:

1. Dust-like component (candidate dark-matter analog):

$$\rho_{\text{dust}} = \frac{1}{20}S, \quad p_{\text{dust}} = 0.$$

1. Negative-pressure component (candidate dark-energy analog):

$$\rho_{\text{neg}} = \frac{19}{20}S^{1/5}, \quad p_{\text{neg}} = -\frac{4}{5}\rho_{\text{neg}}.$$

1. Total equation-of-state trajectory:

$$w = \frac{p}{\rho},$$

which evolves from less negative to more negative effective pressure dominance as S dilutes.

1. Fraction diagnostics used by the solver output:

$$f_{\text{dust}} = \frac{(1/20)S}{\rho}, \quad f_{\text{neg}} = \frac{(19/20)S^{1/5}}{\rho}.$$

Interpretation guardrails:

1. These are model-internal effective-fluid analogies, not direct observational identifications.
2. Split signature $(4,4)$ is mathematically consistent in this framework but is not a standard Lorentzian cosmology.
3. The report does not claim particle-physics identity, perturbation stability, or precision-fit cosmology.

6 Reproduction commands

6.1 Windows PowerShell

```
.\scripts\verify_phase7_x0_x7_reports.ps1
```

6.2 Git Bash or WSL

```
bash ./scripts/verify_phase7_x0_x7_reports.sh
```